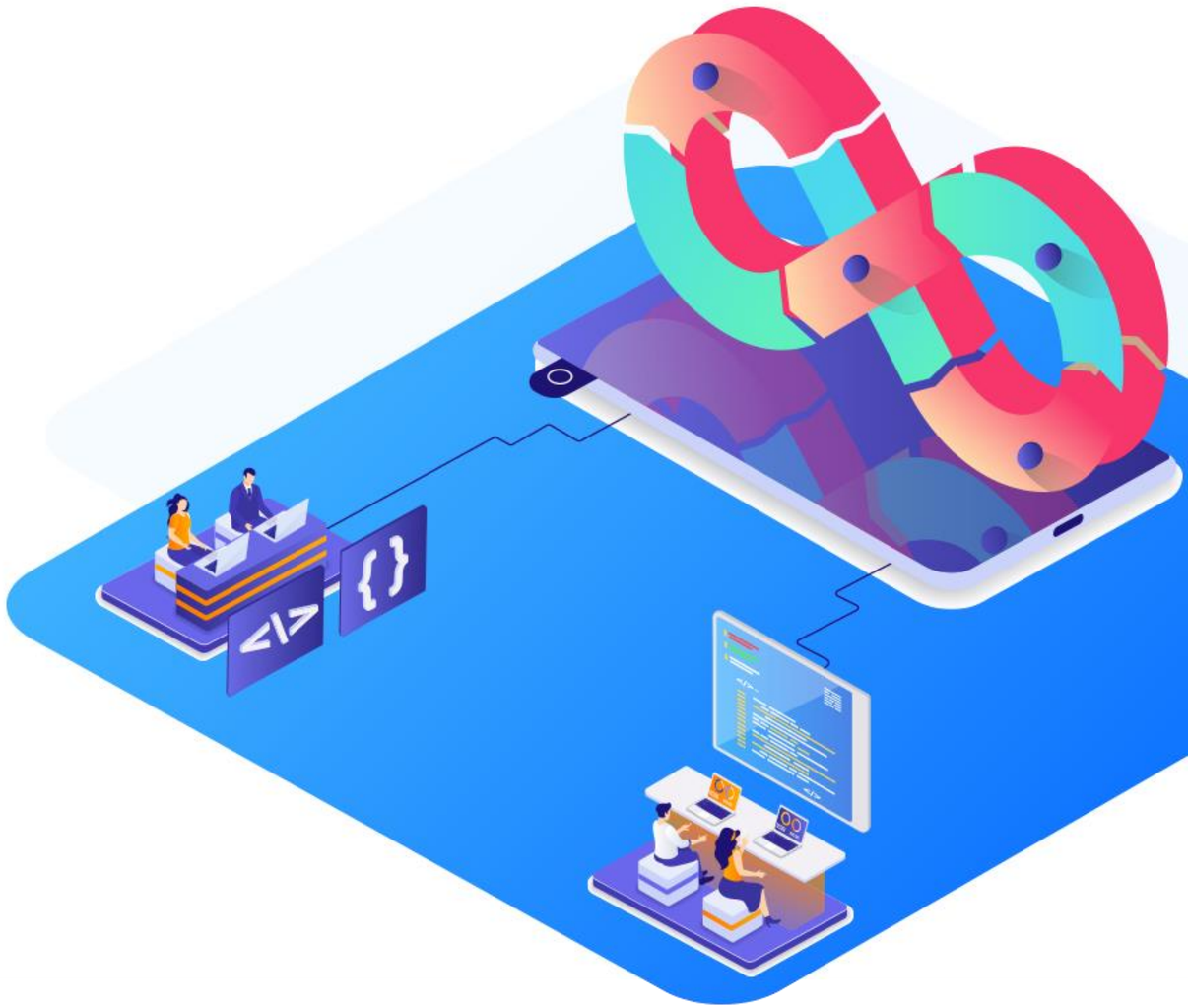
Configuration Management with Ansible and Terraform



Working with Ansible Roles and Vault

Learning Objectives

By the end of this lesson, you will be able to:

- Implement Ansible roles in a playbook to organize and reuse configurations
- Configure the Ansible roles directory structure to manage multiple roles efficiently
- Create a personalized collection of roles on Ansible Galaxy to standardize configurations across projects
- Create a secure playbook using Ansible Vault to manage sensitive information like passwords and keys

Ansible Roles

What Is an Ansible Role?

It is a self-contained and portable unit of automation that groups related tasks and associated variables, files, handlers, and other assets in a known file structure.

ANSIBLE

It can create bundles of automation content that you can run in one or more plays, reuse across playbooks, and share with other users in collections.

What Is an Ansible Role?

Key facts about Ansible roles:

01 They must be used within a playbook.

02 They are defined using YAML files with a predefined directory structure.

03 They are sets of tasks that configure a host to serve a certain purpose.

What Is an Ansible Role?

04 They enhance code reusability by providing predefined configurations and tasks.

05 They can be easily modified due to their structured format.

Why Use an Ansible Role?

Below are the key reasons for using Ansible roles with Ansible playbook:

Reusability
Roles are reusable across multiple plays

Modularity
Each role performs a specific functionality

Organization
Roles provide a structured way to organize tasks

Parametrization
Roles provide overridable default variables

Dependency management
Roles can specify dependencies

Versioning
Roles can be versioned independently

Simplicity
Roles simplify the use of playbooks

Ansible Roles: Use Cases

Ansible roles offer a way to group tasks, handlers, variables, files, templates, and modules to automate IT processes in a structured and reusable manner. Below are some of its use cases:

Server provisioning and configuration	Automate the setup of new servers to ensure consistency and reduce manual errors
Application deployment	Deploy applications across different environments (development, staging, production) with consistency

Ansible Roles: Use Cases

Security and compliance	Ensure that systems comply with security policies and standards
Patch management	Automate the process of applying patches and updates to systems to maintain security and performance
Disaster recovery and backup	Automate backup and recovery processes to ensure data integrity and availability in case of a disaster

Quick Check

You are working on an automation project and need to organize your Ansible playbooks for better reusability and maintainability. Which of the following actions best demonstrates the use of an Ansible role in this scenario?

A. Writing a single, large playbook that includes all tasks and variables

B. Creating a custom module to handle specific tasks within the playbook

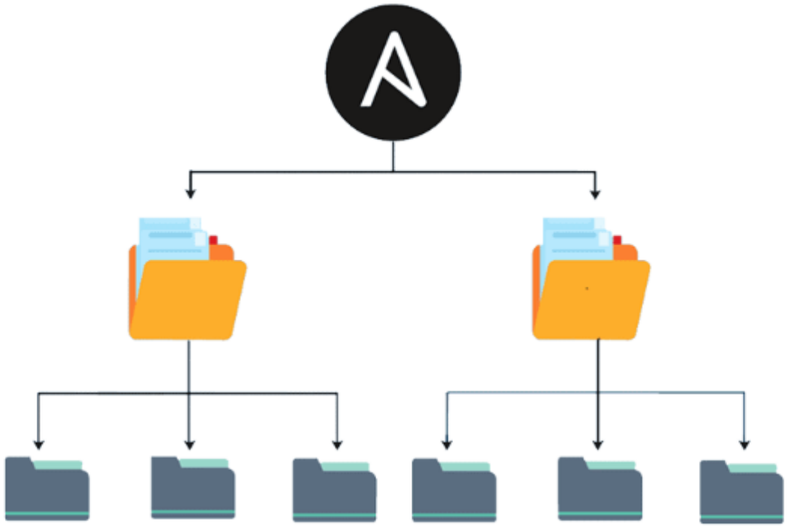
C. Grouping related tasks, variables, and handlers into a structured directory and referencing them in your playbooks

D. Using Ansible Tower to manage and execute your playbooks

Ansible Role Directory

What Is an Ansible Role Directory?

An Ansible role directory is a predefined directory structure that helps organize various components of a role, such as tasks, handlers, variables, templates, and files.



Each role must include at least one of the seven standard directories, and unused directories can be omitted.

Source: https://k21academy.com/ansible/roles/

Ansible Role Directory Structure

The following are the seven standard directories and files:

tasks/main.yml

Lists tasks that the role provides to the play for execution

handlers/main.yml

Imports handlers into the parent play for use by the role or other roles and tasks in the play

defaults/main.yml

Contains default variables for the role which can be overridden

vars/main.yml

Stores variables that have higher precedence than those in **defaults/main.yml**

Ansible Role Directory Structure

files/stuff.txt

Contains one or more files that are available for the role and its children

meta/main.yml

Contains metadata for the role, including role dependencies on other roles

templates/something.j2

Stores Jinja2 templates that can be used to generate configuration files dynamically

By default, Ansible searches for a **main.yml** file in most of the role directories to find the relevant content.

Ansible Role Directory Structure

Example:

```
roles/
    common/                 # this hierarchy represents a "role"
        tasks/              #
            main.yml        #  <-- tasks file can include smaller files if
warranted
        handlers/           #
            main.yml        #  <-- handlers file
        templates/          #  <-- files for use with the template resource
            ntp.conf.j2     #  <------- templates end in .j2
        files/              #
            bar.txt         #  <-- files for use with the copy resource
            foo.sh          #  <-- script files for use with the script resource
        vars/               #
            main.yml        #  <-- variables associated with this role
        defaults/           #
            main.yml        #  <-- default lower priority variables for this role
        meta/               #
            main.yml        #  <-- role dependencies
        library/            # roles can also include custom modules
        module_utils/       # roles can also include custom module_utils
        lookup_plugins/     # or other types of plugins, like lookup in this case
```

Quick Check

You are tasked with creating an Ansible role for setting up a web server. Which of the following directory structures correctly represents the seven standard directories in an Ansible role?

A. bin, etc, lib, opt, sbin, usr, var

B. handlers, tasks, templates, files, vars, defaults, meta

C. scripts, configs, logs, data, vars, handlers, meta

D. docs, src, build, tests, configs, roles, tasks

Creating an Ansible Role

Creating an Ansible Role

The following command will create a new role in the current directory:

Syntax

```
ansible-galaxy init role_name
```

To create a role within a specified directory, use the following syntax:

Syntax

```
ansible-galaxy init directory_path/role_name
```

This command will build a directory that follows the standard role directory structure. Once the role directory is created, you can define tasks, default variables, and other components of an Ansible role.

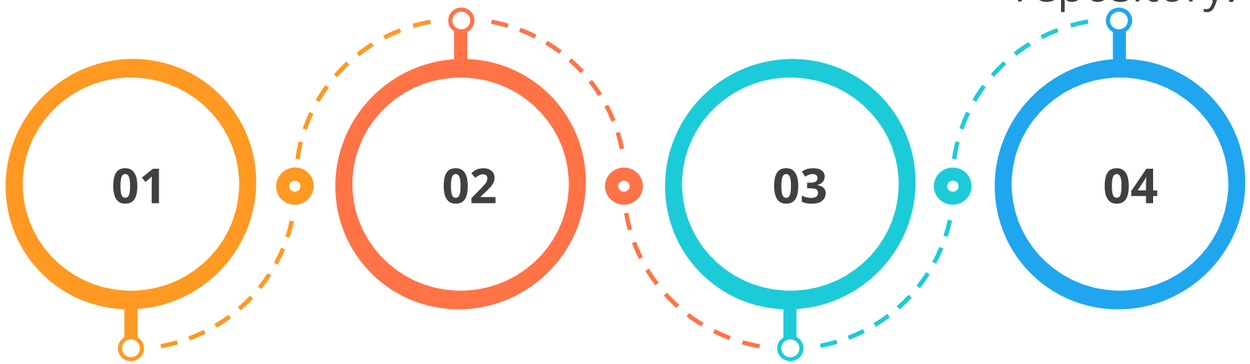
Example: Creating an Ansible Role

```
ansible-galaxy init /etc/ansible/roles/myrole --offline
```

The following steps occur when the above command is executed for creating a role:

init is used to initialize the role.

Create **offline** mode rather than fetching from online repository.



ansible-galaxy is the command to create roles using the templates.

myrole is the name of the role.

Tasks in an Ansible Role

A task in an Ansible role is a discrete action that Ansible performs on a managed node. Below are the steps to specify tasks within an Ansible role:

Navigate to the role directory. For example, **cd /etc/ansible/roles/myrole**

Locate the **tasks** directory in which there will be a file named **main.yml** to define the tasks

Edit the **main.yml** file to define tasks. For example, **nano tasks/main.yml**

Tasks in an Ansible Role

Here is an example **tasks/main.yml** file for installing and starting the Apache web server:

```
---
# tasks file for myrole

- name: Install Apache
  apt:
    name: apache2
    state: present
  become: yes

- name: Start Apache service
  service:
    name: apache2
    state: started
    enabled: yes
  become: yes
```

Storing and Finding a Role

Ansible roles are stored in several default locations. These include:

- The **roles/** directory relative to the playbook file
- The configured **roles_path**
- The default search path, including **~/.ansible/roles, /usr/share/ansible/roles** and **/etc/ansible/roles**
- The directory where the playbook file is located

If the role is stored in a different location, the role path option must be set so that Ansible can find the roles.

Using Ansible Roles

The following are the ways to use an Ansible role:

01 At the **play level** with the roles option

02 At the tasks level with **include_role**

03 At the tasks level with **import_role**

04 As a dependency on another role

Using Ansible Roles at Play Level

Ansible roles are used in a play using the **roles** option:

Example:

```
---
- hosts: webservers
  roles:
    - common
    - webservers
```

This code snippet demonstrates how to apply multiple roles (**common** and **webservers**) to the hosts specified in the **webservers** group.

Using Ansible Roles at Play Level

When the **roles** option is used at a play level, each role **x** looks for a **main.yml** (also **main.yaml** and **main**) in the following directories:

If **roles/x/tasks/main.yml** exists	Ansible adds the tasks in that file to the play
If **roles/x/handlers/main.yml** exists	Ansible adds the handlers in that file to the play
If **roles/x/vars/main.yml** exists	Ansible adds the variables in that file to the play
If **roles/x/defaults/main.yml** exists	Ansible adds the variables in that file to the play
If **roles/x/meta/main.yml** exists	Ansible adds any role dependencies in that file to the list of roles

Additionally, any copy, script, template, or include tasks (in the role) can reference files in **roles/x/{files, templates, tasks}/** without having to path them relatively or absolutely.

Using Ansible Roles at Play Level

Ansible executes a play using the **roles** option at the play level in the following sequence:

01 Executes any **pre_tasks** defined in the play

02 Runs any handlers triggered by **pre_tasks**

03 Processes each role listed in the **roles:** section, in the specified order

Using Ansible Roles at Play Level

04 Executes any tasks defined in the play

05 Runs any handlers triggered by the roles or tasks

06 Executes any **post_tasks** defined in the play

07 Runs any handlers triggered by **post_tasks**

Using Ansible Roles Dynamically

Users can reuse roles dynamically anywhere in the **tasks** section of a play using **include_role**.

01 The included roles run in the order they are defined.

02 If there are other tasks before an **include_role** task, the other tasks will run first.

Example: Using Ansible Roles Dynamically

```
---

-  hosts: webservers
     tasks:
       - name: Print a message
         ansible.builtin.debug:
             msg: "this task runs before the example role"

       - name: Include the example role
          include_role:
             name: example

       - name: Print a message
         ansible.builtin.debug:
              msg: “this task runs after the example role”
```

Example: Using Ansible Roles Dynamically

The following shows that keywords, such as variables and tags, can be included while using roles in a play:

```
---
- hosts: webservers
  tasks:
    - name: Include the foo_app_instance role
      include_role:
        name: foo_app_instance
      vars:
        dir: '/opt/a'
        app_port: 5000
      tags: typeA
  ...
```

Ansible applies the tag only to the **include_role** section which allows the execution of the selected tasks from the role.

Example: Using Ansible Roles Dynamically

The below example shows how to add Ansible roles dynamically using conditions:

```
---
- hosts: webservers
  tasks:
    - name: Include the some_role role
      include_role:
        name: some_role
      when: "ansible_facts['os_family'] == 'RedHat'"
```

Why Use Dynamic Ansible Roles

Below are the advantages of using dynamic Ansible roles using **include_role**:

Flexibility

Flexible and conditional execution of tasks

Scalability

Easily scale automation by reusing roles

Version control

Management of different versions of roles

This approach enhances the modularity and maintainability of the automation scripts.

Using Static Ansible Roles

Users can reuse roles statically anywhere in the tasks section of a play using **import_role**.

01 Roles are parsed and included during playbook parsing.

02 The tasks from the imported roles are executed in the defined sequence.

Note:

When a tag is added to an **import_role** statement, Ansible applies the tag to all tasks within the role.

Example: Using Ansible Roles Statically

```
---

 - hosts: webservers
    tasks:
         - name: Print a message
            ansible.builtin.debug:
                  msg: "before we run our role"

          - name: Import the example role
             import_role:
                  name: example

          - name: Print a message
             ansible.builtin.debug:
                    msg: "after we ran our role"
```

Why Use Static Ansible Roles?

Below are the advantages of using static Ansible roles using **import_role**:

Portability

Ensures uniform execution in different environments

Isolation

Ensures isolated execution of roles in a predictable order

Version control

Allows for better version control

Performance

Reduces execution time and increases efficiency

This approach enhances the consistency and reliability of the automation scripts.

Multiple Execution of a Role

Running a role multiple times in one playbook

Ansible executes each role once, even if it is defined multiple times.

Roles are executed multiple times only if the parameters defined on the role are different for each definition.

Ansible only runs the role **xyz** once in this play.

Example

```
---
-  hosts: webservers
        roles:
            - xyz
            - abc
            - xyz
```

Multiple Execution of a Role

There are two options to force Ansible to run a role more than once:

01 Passing different parameters

02 Using the **allow_duplicates: true** parameter

Multiple Execution of a Role

Difference between the two options to force multiple Ansible role execution:

Passing different parameters	Allowing duplicates
Execute the role multiple times by passing different sets of parameters	Execute the role multiple times with the same set of parameters by explicitly allowing duplicates
Perform different tasks or configurations based on parameters	Re-apply the same role without changing parameters
Include or import the role multiple times, each with different variables	Include or import the role multiple times with **allow_duplicates: true**

Example: Passing Different Parameters

This playbook runs the **xyz** role twice:

Example:

```
---
-  hosts: webservers
       roles:
           role: xyz
            message: "first"

           role: xyz
             message: "second"
```

Ansible runs **xyz** twice because each role definition has different parameters.

Example: Allow Duplicates

The following configuration executes the same role multiple times by explicitly enabling duplicates:

Example

```
# playbook.yml
---
- hosts: webservers
  roles:
   - foo
   - foo
# roles/foo/meta/main.yml
---
allow_duplicates: true
```

Using **allow_duplicates: true** ensures that specific roles are re-applied as needed, providing flexibility and control in complex playbook executions.

Assisted Practice

Creating Ansible roles **Duration: 20 Min.**

Problem statement:

You have been assigned a task to create and configure Ansible roles for managing configurations.

Outcome:

By the end of this demo, you will be able to write Ansible roles to organize playbooks into reusable, modular components, making it easier to manage complex configurations by encapsulating related tasks, variables, files, templates, and handlers into structured directories.

Note: Refer to the demo document for detailed steps

Assisted Practice: Guidelines

Steps to be followed:

1. Initialize the roles
2. Create and define tasks
3. Organize task execution
4. Create the index.html file
5. Verify the configuration
6. Execute the playbook
7. Check the index.html file in the browser

ASSISTED PRACTICE

Quick Check

You need to reuse an Ansible role called **webserver** in your playbook, ensuring that it runs multiple times with different configurations. Which approach allows you to achieve this dynamically at the task level?

A. Including the **webserver** role in the roles section of the playbook with default configurations

B. Using the **include_role** module within tasks and passing different variables for each execution

C. Defining multiple **webserver** tasks statically in the tasks section with different variables

D. Creating multiple identical playbooks each specifying the **webserver** role with different configurations

Ansible Galaxy

What Is Ansible Galaxy?

It is a community hub for finding, sharing, and downloading Ansible content, including roles and collections. It provides pre-packaged units of work that help streamline and enhance automation projects.

GALAXY

Users can search, install, and manage roles and collections using the **ansible-galaxy** command-line tool.

Why Use Ansible Galaxy?

Ansible Galaxy serves several key purposes when working with Ansible roles, enhancing the efficiency and effectiveness of automation tasks:

1. It provides a vast repository of pre-built roles contributed by the community.
2. It provides roles that are well-documented and follow best practices, ensuring standardization.
3. It allows users to share their roles with the community, making it easier to collaborate.
4. It provides roles which makes the playbook modular and manageable.

Common Ansible Galaxy Commands

The following are some of the most common Ansible Galaxy commands:

`ansible-galaxy list`	To display the list of installed roles with version numbers
`ansible-galaxy remove [role]`	To remove an installed role
`ansible-galaxy init`	To create a role template suitable for submission to Ansible Galaxy

Galaxy: Collection

Collections are a distribution format for Ansible content that includes roles, modules, plugins, and other resources.

It can be installed and utilized through a distribution server, such as Ansible Galaxy or a Pulp 3 Galaxy server.

Source: https://www.ansible.com/blog/migrating-to-ansible-collections-webinar-qa

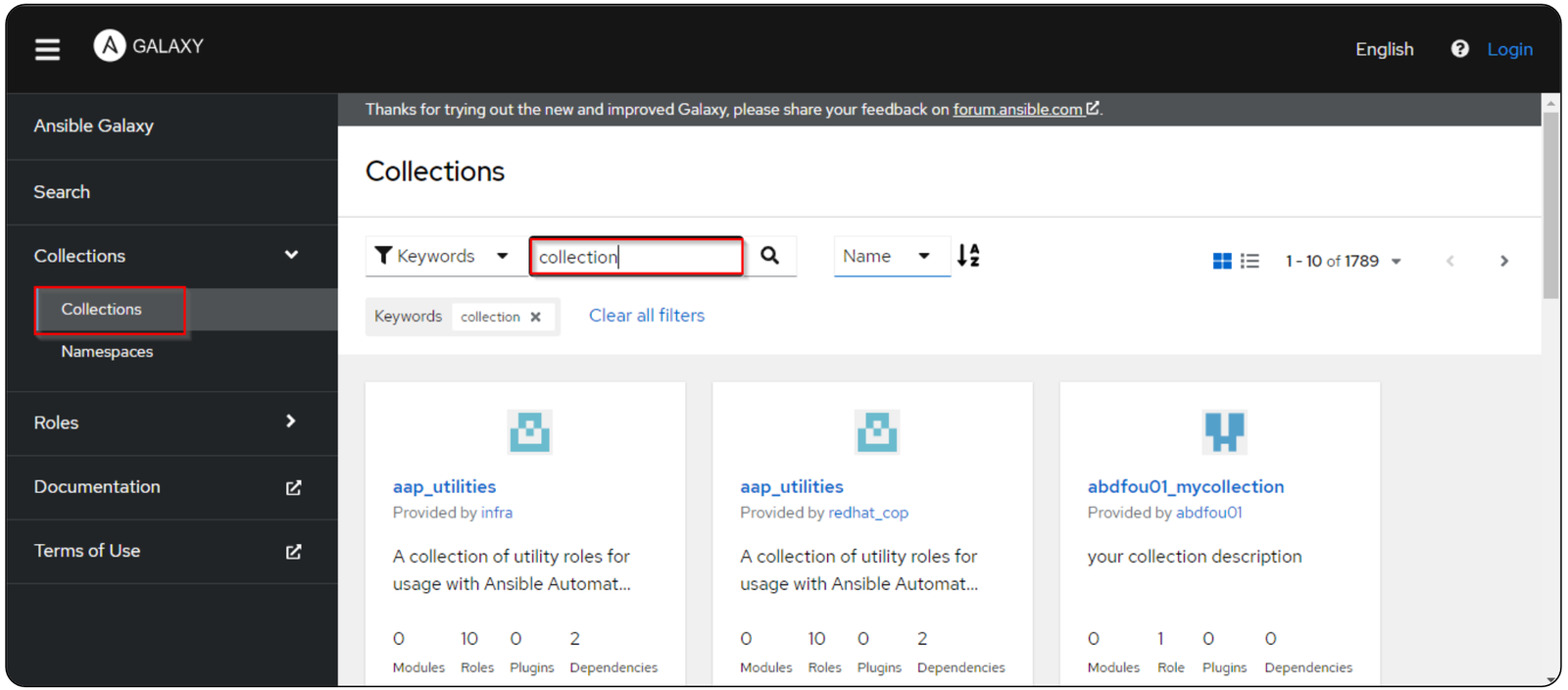
Galaxy: Collection

To find collections on the Ansible Galaxy console:

Click on the **Search** icon in the left-hand navigation

Set the filter to the **collection**

Set the other filters and press **Enter**



Galaxy: Collection Structure

A collection structure refers to the organized layout of directories and files that make up an Ansible collection.

01 A collection requires a **galaxy.yml** file at the root level of the collection.

02 This file contains all the metadata that Galaxy and other tools need to package, build, and publish the collection.

Galaxy: Collection Structure

A collection follows the following simple structure of directories and files:

```
collection/
├── docs/
├── galaxy.yml
├── plugins/
│ ├── modules/
│ │ └── module1.py
│ ├── inventory/
│ └── .../
├── README.md
├── roles/
│ ├── role1/
│ ├── role2/
│ └── .../
├── playbooks/
│ ├── files/
│ ├── vars/
│ ├── templates/
│ └── tasks/
└── tests/
```

Installing a Collection from Galaxy

The following command is used to install a collection from Ansible Galaxy using the **ansible-galaxy** command-line tool:

```
ansible-galaxy collection install collection_name
```

The command by default uses **https://galaxy.ansible.com** as the Galaxy server.

To upgrade a collection to the latest available version from the Galaxy server, users can use the **--upgrade** option after the collection name.

Quick Check

You need to install multiple roles and collections in your Ansible project, ensuring that all dependencies are managed automatically. Which file format and command should you use to achieve this?

A. Create a **requirements.txt** file and use **ansible-galaxy install -r requirements.txt**

B. Create a **roles.yml** file and use **ansible-galaxy install -r roles.yml**

C. Create a **requirements.yml** file and use **ansible-galaxy install -r requirements.yml**

D. Create a **collections.yml** file and use **ansible-galaxy install -r collections.yml**

Ansible Vault

What Is Ansible Vault?

It is a feature of Ansible that provides encryption for sensitive data, such as passwords, keys, and other confidential information.

It integrates with third-party key management services like Amazon AWS KMS, enabling secure cloud management and storage of encrypted secrets.

Source: https://www.educba.com/ansible-vault/

Operations Performed by Ansible Vault

Encrypt a file

Decrypt a file

View an encrypted file

Edit an encrypted file

Create an encrypted file

Generate or reset an encrypted file

Encryption Levels in Ansible

Ansible vault provide following levels of encryption:

File-level encryption

Ansible Vault can encrypt any structured data file used by Ansible, including inventory variables, role variables, task files, handlers, and arbitrary file

Variable-level encryption

Ansible Vault can encrypt single values inside a YAML file using the **!vault** tag to indicate special processing for those values.

Encryption with Ansible Vault

Below are the steps to encrypt or decrypt content with Ansible Vault:

01 Access passwords stored in a third-party tool using a script

02 Encrypt or decrypt content using the ansible-vault command-line tool with passwords

03 Store encrypted content safely in source control and share it securely

Encryption with Ansible Vault

Encrypted variables and files can be used in ad hoc commands and playbooks by providing the relevant passwords used during encryption.

ansible.cfg

Users have the option to adjust the **ansible.cfg** file to specify the location of a password file or to set it to always prompt for the password.

Encryption with Ansible Vault

Users can encrypt variables and files with Ansible Vault using:

!vault	This tag tells Ansible and YAML that the content needs to be decrypted
|	This character allows multi-line strings
--vault-id	This label specifies which vault ID decrypts the content

Encrypting Individual Variables with Ansible Vault

The **ansible-vault encrypt_string** command used to create a basic encrypted variable.

The options passed in the command are:

01 A source for the vault password

02 The string to encrypt and the string name

Syntax:

```
ansible-vault encrypt_string <password_source> '<string_to_encrypt>' --name '<string_name_of_variable>'
```

Encrypting Individual Variables with Ansible Vault

The following command is used to encrypt the string **samplestring** using the password stored in **a_password_file**:

Example:

```
ansible-vault encrypt_string --vault-password-file a_password_file
'samplestring' --name 'the_secret'
```

Output:

```
the_secret: !vault |
      $ANSIBLE_VAULT;1.1;AES256
      62313365396662343061393464336163383764373764613633653634306231386433626436623361
      61343336653539663635343336326665353337616661316 20a66353764643664383961653164356 1
      63396265333966386166373632626539326166353965363262633030333630313338646335303630
      34386266666661376 50a353638643435666633633964366338633066623234616432373231333331
      6564
```

Encrypting Individual Variables with Ansible Vault

The user can view the original value of the encrypted variable in the debug module.

Example:

```
ansible localhost -m ansible.builtin.debug -a var="the_secret" -e "@vars.yml" --vault-id dev@a_password_file
```

Output:

```
localhost | SUCCESS => {
    "the_secret": "samplestring"
}
```

Encrypting Files with Ansible Vault

It can encrypt any structured data file used by Ansible. The full file is encrypted in the vault. Here are some key types of files that Ansible Vault can encrypt:

- Group variables files from inventory
- Variables files in roles
- Tasks and handler files
- Variables files loaded by **include_vars** or **vars_files**

Encrypting Files with Ansible Vault

Here are some key types of files that Ansible Vault can encrypt:

- Host variable files from inventory
- Default files in roles
- Binary files or arbitrary files
- Variables files passed to ansible-playbook with -e @file.yml or -e @file.json

How to Encrypt Files with Ansible Vault

To create a new encrypted data file named **sample.yml** using the test vault password from **multi_password_file**, execute the following command:

```
ansible-vault create --vault-id test@multi_password_file sample.yml
```

Executing this command opens an editor to add content. Upon closing the editor, the file is saved as encrypted content. The file header shown below displays the vault ID used for its creation:

```
``$ANSIBLE_VAULT;1.2;AES256;test``
```

How to Encrypt Files with Ansible Vault

The command to encrypt an existing file is **ansible-vault encrypt**. It can operate on multiple files as shown below:

```
ansible-vault encrypt foo.yml bar.yml baz.yml
```

The command to view the contents of an encrypted file without editing, is the **ansible-vault view.** It can display multiple files as shown below:

```
ansible-vault view foo.yml bar.yml baz.yml
```

Ansible Vault Commands

The commonly used commands while encrypting files using Ansible Vault are:

ansible-vault edit foo.yml	Modifies an encrypted file in place without needing to manually decrypt and re-encrypt it
ansible-vault rekey foo.yml	Changes the encryption password of the specified file, enhancing security without altering the file content
ansible-vault decrypt foo.yml	Decrypts an encrypted file, making it accessible for editing or inspection without encryption

Format of Files Encrypted with Ansible Vault

Ansible Vault creates Unicode Transformation Format (UTF-8) encoded txt files:

Example:

$ANSIBLE_VAULT;1.1;AES256

or

$ANSIBLE_VAULT;1.X;AES256;vault-id-label

The file format includes a header terminated by a newline. This header comprises up to four elements separated by semicolons.

Assisted Practice

Implementing Ansible Vault **Duration: 20 Min.**

Problem statement:

You have been assigned a task to securely manage sensitive files using Ansible Vault, ensuring encryption and protection of sensitive data.

Outcome:

By the end of this demo, you will be able to implement Ansible Vault to securely store sensitive data such as passwords, API keys, and other confidential information within Ansible playbooks.

Note: Refer to the demo document for detailed steps

Assisted Practice: Guidelines

Steps to be followed:

1. Create and edit a vault
2. Reset the password for the vault
3. Execute the YAML file

ASSISTED PRACTICE

Quick Check

You are tasked with securely storing sensitive information, such as API keys and passwords, within your Ansible playbooks. You decide to use Ansible Vault to encrypt this data. Which Ansible Vault command is used to encrypt a file containing sensitive information?

A. ansible-vault lock

B. ansible-vault hide

C. ansible-vault encrypt

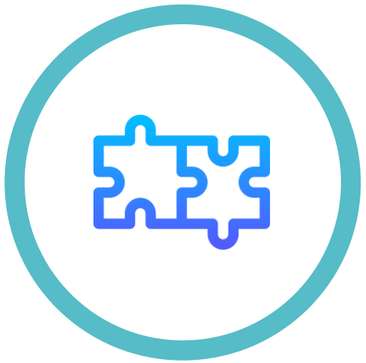
D. ansible-vault secure

Working with Dynamic Inventory

What Is Dynamic Inventory?

It is a feature in automation tools like Ansible that automatically retrieves and manages inventory data from various sources in real time.

- It provides information about public and private cloud providers, cobbler system information, and CMDB (Configuration Management Database).
- Inventory plugins are preferred over scripts because they align with the latest updates in the Ansible core code and offer more customization options.



Inventory Plugins

Inventory Scripts

Dynamic Inventory Plugins

Inventory plugins allow users to connect to various data sources to generate the host inventory that Ansible uses to target jobs.

Points to Remember:

- The inventory plugins bundled with Ansible are enabled by default or can be used with the **auto** plugin.
- The **ansible.cfg** configuration file can override the default list of enabled plugins. Below is the default list of enabled plugins that ship with Ansible:

  [inventory] enable_plugins = host_list, script, auto, yaml, ini, toml

Dynamic Inventory Scripts

Ansible uses inventory scripts to fetch host information from external inventory systems.

```
{
    "group001": {
        "hosts": ["host001", "host002"],
        "vars": {
            "var1": true
        },
        "children": ["group002"]
    },
    "group002": {
        "hosts": ["host003", "host004"],
        "vars": {
            "var2": 500
        },
        "children": []
    }
}
```

Inventory scripts must accept **--list** and **--host <hostname>** arguments, returning detailed information about host groups and individual hosts.

Assisted Practice

Implementing dynamic inventories **Duration: 20 Min.**

Problem statement:

You have been assigned a task to demonstrate how to work with dynamic inventories in Ansible, enabling the automatic fetching and usage of inventory data from external sources.

Outcome:

By the end of this demo, you will be able to implement Ansible Vault to securely store sensitive data such as passwords, API keys, and other confidential information within Ansible playbooks.

Note: Refer to the demo document for detailed steps

Assisted Practice: Guidelines

Steps to be followed:

1. Install the required tools
2. Create and configure a dynamic inventory script
3. Create and execute a playbook using the dynamic inventory script

ASSISTED PRACTICE

Quick Check

You are managing a cloud infrastructure where servers are frequently added and removed. To keep your Ansible inventory up-to-date without manual intervention, you decide to use a dynamic inventory. Which feature of Ansible allows you to automatically update your inventory based on the current state of your cloud infrastructure?

A. Static inventory

B. Inventory groups

C. Dynamic inventory plugins

D. Inventory variables

Key Takeaways

- Roles are not playbooks. They are small functions that can be used individually within playbooks.
- Ansible Galaxy is a free Galaxy website for finding, downloading, and sharing community-developed roles and collections.
- Collections can be used to package and distribute roles, modules, playbooks, and plugins.
- Roles can be dependent on other roles, and when a user installs a role that has dependencies, those dependencies will automatically be installed to the **roles_path**.

Key Takeaways

- Ansible Vault is a tool designed to encrypt confidential information, allowing users to safeguard sensitive data like passwords from being exposed in plain text.
- Ansible Vault has the capability to encrypt any structured data file utilized within Ansible. The entire file is encrypted within the vault for added security.

Initializing Ansible Roles and Inventory

Duration: 20 Min.

Project agenda: To automate the deployment and management of an Apache web server using Ansible roles, securely handle sensitive information using Ansible Vault, and dynamically manage the inventory using a custom Python script for AWS EC2 instances

Description: As a DevOps engineer at a growing tech company, you are tasked with automating the deployment of Apache web servers. The company's infrastructure is dynamic, requiring frequent updates and the ability to manage sensitive configuration data securely. Your objective is to create an Ansible role for installing and configuring Apache, use Ansible Vault to manage sensitive data like passwords and implement a dynamic inventory to handle the constantly changing server environment.

Initializing Ansible Roles and Inventory

Duration: 25 Min.

Perform the following:

1. Initialize the Ansible role for Apache
2. Define tasks for the Ansible role
3. Configure and use Ansible Vault
4. Implement dynamic inventory

Expected deliverables: An Ansible playbook that automates Apache web server deployment and management, complete with detailed installation and execution documentation.

Thank you